1.1. Ground Control Station

1.1.1. Overview

The GCS has a portable form, contained in a ruggedized case for rapid and easy transport. However, the GCS also has a configuration that is a permanent or semi-permanent installation in a building.

The datalink consists of a ground radio unit and its antennas. For extended range, the antennas are mounted on a pan-tilt tracker unit that points the antennas at the aircraft to take advantage of high gain directional antennas.

GSS is based on Human Machine Interface (HMI), which is the mission management system developed by Intra Defense Technologies for unmanned aircraft. This interface provides infrastructure and common functions that are required within tactical mission management systems with adaptation and customizations depending on the project specific requirements.

1.1.2. Capabilities

The following tasks are possible using the GCS:

- Authentication and authorization
- Flight planning on map and loading onto the UAV's mission computer
- Tracking UAV on moving map
- Visualization of flight and mission data
- Monitoring subsystems' status
- Displaying and listing alerts, warnings, and information conditions
- Generation of mission commands
- Mission checklist management
- Data link control
- Mission data recording
- Payload control and management

1.1.3. Mobile Ground Control Station (MGCS)



Figure 1 Ground Control Station (portable version)

GCS Case

- X
- Size: 560 mm x 450 mm x 265 mm (closed)
- Housing Material: PP
- IP Rating: IP67 (closed)
- Working Environment: -20°C to 60°C
- Color: Matt Black
- Portable Features: Open and play, three proof delivery, With handle.
- battery & Working Time: Smart battery management system, 9 hrs endurance.
- Battery charges for 5 hours (could be operated while charging)
- Transmitter Solution with Tracking Antenna
- Ground terminal
- Air terminal
- Tracking gimbal
- Gimbal tripod
- Panel antenna